



Review

The origins and evolution of “controlled” drug delivery systems

Allan S. Hoffman

Bioengineering Department, University of Washington, Seattle, WA 98195 USA

ARTICLE INFO

Article history:

Received 15 June 2008

Accepted 18 August 2008

Available online 28 August 2008

Keywords:

Controlled drug delivery history

Alex Zaffaroni and Alza

Judah Folkman

Nanocarriers

PEGylation

Targeted drug delivery

ABSTRACT

This paper describes the earliest days when the “controlled drug delivery” (CDD) field began, the pioneers who launched this exciting and important field, and the key people who came after them. It traces the evolution of the field from its origins in the 1960s to (a) the 1970s and 1980s, when numerous *macroscopic* “controlled” drug delivery (DD) devices and implants were designed for delivery as mucosal inserts (e.g., in the eye or vagina), as implants (e.g., sub-cutaneous or intra-muscular), as ingestible capsules (e.g., in the G-I tract), as topical patches (e.g., on the skin), and were approved for clinical use, to (b) the 1980s and 1990s when *microscopic* degradable polymer depot DD systems (DDS) were commercialized, and to (c) the currently very active and exciting *nanoscopic* era of targeted nano-carriers, in a sense bringing to life Ehrlich’s imagined concept of the “Magic Bullet”. The nanoscopic era began with systems proposed in the 1970s, that were first used in the clinic in the 1980s, and which came of age in the 1990s, and which are presently evolving into many exciting and clinically successful products in the 2000s. Most of these have succeeded because of the emergence of three key technologies: (1) PEGylation, (2) active targeting to specific cells by ligands conjugated to the DDS, or passive targeting to solid tumors via the EPR effect.

The author has been personally involved in the origins and evolution of this field for the past 38 years (see below), and this review includes information that was provided to him by many researchers in this field about the history of various developments. Thus, this paper is based on his own personal involvements in the CDD field, along with many historical anecdotes provided by the key pioneers and researchers in the field. Because of the huge literature of scientific papers on CDD systems, this article attempts to limit examples to those that have been *approved for clinical use, or are currently in clinical trials*. Even so, it is impossible to know of and include all such examples and to properly credit all the key people who helped to bring the various technologies and devices to the clinic. The author apologizes in advance for all omissions.

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1. The origins of “controlled” drug delivery

In the mid-1960s Judah Folkman, MD, at Harvard was circulating rabbit blood inside a Silastic® (silicone rubber) arterio-venous shunt and discovered that if he exposed the tubing to anaesthetic gases on

the outside, the rabbits would fall asleep. [1] He proposed that short, sealed segments of such tubing containing a drug could be implanted, and if the silicone didn’t change in dimensions or composition, the implant would become a *constant rate* drug delivery device. [2] (Fig. 1) He also showed that the rate decreased as the tubing thickness increased, which is obvious today, but back then it was the first suggestion of a zero order *controlled* drug delivery implant *in vivo*. Meanwhile, across the country in Palo Alto CA, Alejandro (“Alex”)

E-mail address: hoffman@u.washington.edu.

The Use of Silicone Rubber as a Carrier for Prolonged Drug Therapy

JUDAH FOLKMAN, M.D., AND DAVID M. LONG, LCDR, MC, USNR, *National Naval Medical Center*



Folkman

SUMMARY

A new method is presented for local or systemic drug therapy by incorporating the therapeutic agent in a Silastic capsule which is then implanted in the body.

J Surgical Research, 4, 139 (1964)

Fig. 1. The 1964 article in which Judah Folkman proposed the original concept of an implanted, zero-order, reservoir drug delivery device. (See text for details).

Zaffaroni, an outstanding synthetic drug chemist and entrepreneur, had been thinking about the concept of zero order delivery and controlled delivery devices. He heard about Folkman's work and went to Boston to learn more about it from him. Stimulated by Folkman's work, as well as by his own vision, Zaffaroni founded a company in the late 1960s focused on the concept of *controlled drug delivery* (CDD). He called it Alza, after the first two letters of his first and last names. He invited Folkman and another drug delivery scientist, Takeuchi ("Tak") Higuchi, at the University of Kansas, either to take top executive positions with Alza, or to be on his Scientific Advisory Board. They both chose to join Alza's SAB. In 1970 he invited Alan Michaels, a former chemical engineering professor from MIT and the founder and President of Amicon Corp., to be President of Pharmetrics, a new company he founded to design and develop the different zero order DDS for Alza. Pharmetrics, which soon merged with Alza, was located a few blocks from Alza headquarters in Palo Alto CA. (Fig. 2) In September of 1970, Michaels, who had personally mentored me when I was a student and faculty member in the Chemical Engineering Department at MIT in the 1950s and 1960s, called me and invited me to become a consultant to Pharmetrics. I visited Alza as a consultant every year since 1970, until 2006, when J&J closed Alza.

2. The "MACRO era" of zero-order "controlled" drug delivery devices

The first CDD devices that Alza designed were macroscopic in scale. One was an ophthalmic insert called the Ocusert® that released the anti-glaucoma drug, pilocarpine, at a constant rate in the eye, and another was called the Progestesert®, an intrauterine device (IUD) that released the contraceptive steroid, progesterone, at a constant rate in the uterine cavity. (Fig. 3) Each used poly(ethylene-co-vinyl acetate) or polyEVA as the rate-controlling membrane (RCM) in a constant rate, reservoir DDS device, producing a zero order, flat PK. Later the Population Council developed a contraceptive subcutaneous implant comprised of 6 silicone rubber tubes (crosslinked polydimethylsiloxane, PDMS) filled with a contraceptive steroid, levo-Norgestrel, which they called the Norplant®. (Fig. 3) This was a direct extension of Folkman and Long's suggestion in their 1964 publication. [2] (Fig. 1) Organon has recently brought a similar device to the clinic (Implanon®), using polyEVA as the RCM. Implanon® was approved for clinical use by the FDA in 2006. Around the same time, in the late

1970s to 1980s, the World Health Organization (WHO) developed a contraceptive doughnut-shaped device for vaginal insertion made of silicone rubber and that used silicone rubber as the RCM; it contained an internal doughnut-shaped core of a contraceptive steroid blended with silicone rubber, and coated with a drug-free PDMS as the RCM. (Fig. 3)

Alza was also developing the controlled delivery skin patch, and in 1971 Zaffaroni was issued one of the first patents on the patch, which he called a "Bandage for Administering Drugs". [3] (Fig. 4) This device is a membrane-controlled reservoir device that exhibits zero order release kinetics. The most commonly used RCMs are polyEVA or porous polypropylene (PP) (the pores could be filled with an oil or wax). The RCM resistance to drug diffusion was designed to be greater than that of the stratum corneum. The first skin patch product contained scopolamine, an anti-motion sickness drug. Alza tested it by chartering a large sailboat for some of its employees, and they sailed under the Golden Gate bridge through a rough patch just north of the bridge called the "potato patch". Some had the drug in their patches and some had placebos. Most of those with the drug-loaded patch didn't get sick while most of those with the placebo did. Zaffaroni was particularly interested in this anti-motion sickness drug patch because he reasoned that with the new space program of President Kennedy, Alza could get government funding to develop such patches for use in space travel.

Beginning in the 1980s and into the 1990s, Alza developed a family of osmotic pump capsules (e.g., Oros®) for controlled release of drugs in the G-I tract, and one of their first patents issued in 1974 [4]. (Fig. 5) Felix Theeuwes, Liang Dong, Guohua Chen, Zhongli Ding and Lothar Kleiner at Alza were major contributors to the development of the various oral and implanted (Duros®) osmotic delivery systems that were under development in the 1990s. The oral capsules also gave a zero order, flat PK within the enteric-colonic tract, and some could be timed to release in certain regions of that tract. Both the skin patch and the oral osmotic capsules are zero order, diffusion-controlled devices with RCMs, but the difference is that, in the patch the RCM controls a constant rate of drug diffusion out of the patch, while in the osmotic capsule the RCM controls a constant rate of water diffusion into the capsule, forcing an equal volume of the drug solution or suspension out of the capsule through a fixed diameter hole. The most commonly used RCM in the osmotic capsule is cellulose acetate, sometimes blended with a small amount of a low MW PEG to get the water diffusion started.

Another zero order DDS for oral delivery of drugs is a cleverly designed swelling/gelling hydrogel system, conceived at the University of Padua in Italy by Paolo Colombo, Ubaldo Conti and

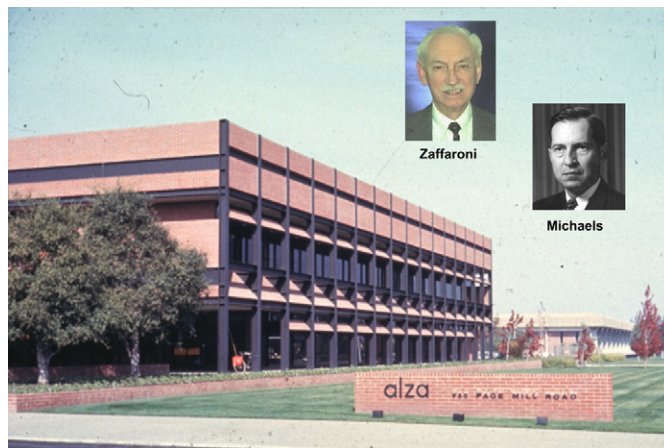


Fig. 2. Photo of Alza Corp. headquarters in Palo Alto, CA. This photo could have been taken in the 1970s. Photos of Alza's founder, Alex Zaffaroni, and his Director of R & D, Alan S. Michaels are shown. (See text for details).

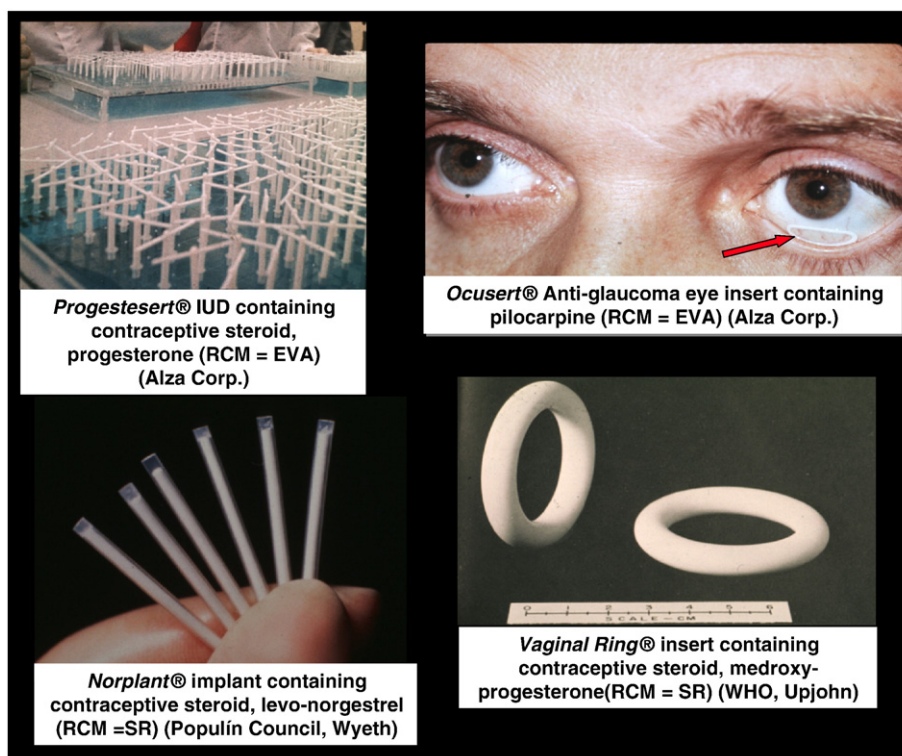


Fig. 3. Photos of the early macroscopic zero-order controlled DDS. The top two were Alza's designed products, and the bottom two came out of the Population Council and World Health Organization. (See text for details).

United States Patent

[11] 3,598,122

[72] Inventor **Alejandro Zaffaroni**
Atherton, Calif.
[21] Appl. No. **812,116**
[22] Filed **Apr. 1, 1969**
[45] Patented **Aug. 10, 1971**
[73] Assignee **Alza Corporation**

3,249,109	5/1966	Maeth et al.	128/268
3,339,546	9/1967	Chen	128/156
3,444,858	5/1969	Russell	128/268 X
3,464,413	9/1969	Goldfarb et al.	128/268
3,518,340	6/1970	Raper	424/19
3,520,949	7/1970	Shepherd et al.	128/156 UX

Primary Examiner—Charles F. Rosenbaum
Attorney—Steven D. Goldby

[54] **BANDAGE FOR ADMINISTERING DRUGS**
15 Claims, 3 Drawing Figs.

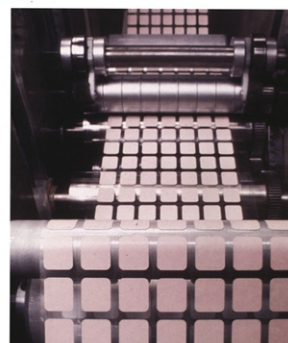
[52] U.S. Cl. **128/268,**
424/20, 424/28
[51] Int. Cl. **A61F 7/02**
[50] Field of Search **128/155-**
— 156, 268, 296; 424/19—20, 28

[56] **References Cited**

UNITED STATES PATENTS

2,629,378	2/1953	Barton	128/268
3,053,255	9/1962	Meyer	128/268

ABSTRACT: Bandage for use in the continuous administration of systemically active drugs by absorption through the skin or oral mucosa comprising a backing member having on one surface thereof a reservoir containing a systemically active drug. The reservoir has a wall distant from the backing member and permeable to passage of the drug. A pressure-sensitive adhesive layer, also permeable to passage of the drug, is carried by the reservoir. The drug is in a form acceptable for absorption through the skin or the mucosa of the mouth.



A rotary die cutting press executes the final manufacturing step for an Alza skin patch.
(photo taken from an Alza annual report)

Fig. 4. One of the earliest (if not the earliest) patents on the skin patch was a 1971 patent issued to Alex Zaffaroni and assigned to Alza.

United States Patent [19] Theeuwes et al.

[11] **3,845,770**[45] **Nov. 5, 1974**[54] **OSMATIC DISPENSING DEVICE FOR
RELEASING BENEFICIAL AGENT**[75] Inventors: **Felix Theeuwes**, Los Altos, Calif.;
Takeru Higuchi, Lawrence, Kans.[73] Assignee: **Alza Corporation**, Palo Alto, Calif.[22] Filed: **June 5, 1972**[21] Appl. No.: **259,469**[52] U.S. Cl.:..... **128/260**, 128/130, 222/389,
424/15, 424/37[51] Int. Cl.:..... **A61m 31/00**[58] Field of Search 128/260, 130, 272, 269,
128/127–129, 213; 99/77.1; 424/15, 37,
19–21; 222/386.5, 389; 206/5[56] **References Cited****UNITED STATES PATENTS**

1,023,499	4/1912	Bell	128/260
1,279,564	9/1918	McClure	128/272
1,434,531	11/1922	Cyrenius	128/272
2,846,057	8/1958	Polin	206/5
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3,146,169	8/1964	Stephenson et al.	424/15
3,416,530	12/1968	Ness	128/260
3,574,820	4/1971	Johnson et al.	424/37 X
3,630,200	12/1971	Higuchi	128/260

Theeuwes

Higuchi

Primary Examiner—Richard A. Gaudet*Assistant Examiner*—J. C. McGowan*Attorney, Agent, or Firm*—Paul L. Sabatine; Edward
L. Mandell; William H. Benz

[57]

ABSTRACT

A wall surrounding and forming a compartment for containing a useful composition of matter and having a passageway for dispensing the composition is disclosed. The wall is comprised in at least a part of a material permeable to an external fluid. The composition is soluble in the fluid and exhibits an osmotic pressure gradient against the fluid or the composition has limited solubility and is admixed with an osmotically effective compound soluble in the fluid that exhibits an osmotic pressure gradient against the fluid. In operation, composition is dispensed from the device by fluid permeating into the compartment producing a solution of the soluble composition or a solution of the osmotically effective compound containing the composition, which solution in either operation is released through the passageway to the exterior of the device at a rate controlled by the permeability of the wall and the osmotic pressure gradient across the wall of the device.

15 Claims, 8 Drawing Figures

Fig. 5. One of the earliest patents on oral osmotic capsules was a 1974 patent issued to Felix Theeuwes and Takeru Higuchi and assigned to Alza.

coworkers and known as the Geomatrix® DDS of Skye Pharma. (REF) Nicholas Peppas also contributed to this very successful oral hydrogel DDS [5].

In the mid-1970s, Folkman, and his former postdoc, Bob Langer (Fig. 7) (who was then an Assistant Professor of Chemical Engineering at MIT), published a seminal paper showing that protein drugs could be gradually released in an active form from macroscopic hydrophobic polymer matrices such as PEVA [6]. Although this device never achieved clinical status, it was significant because it was the first example of successful delivery from a hydrophobic, non-degradable polymer matrix of a “fragile” biomolecular drug such as a protein, and in an active form.

3. The “MICRO era” of sustained release, biodegradable microparticle and phase-separated depot delivery systems

Biodegradable polymers of poly(hydroxy acids) were developed for sutures in the 1960s and 1970s; that technology was adopted by drug delivery researchers in the 1970s and reached the clinic in the 1980s. Thus, the field of controlled DD evolved from the *macroscopic*, zero order PK devices in the 1970s and 1980s, to “sustained release” macro/microscopic biodegradable depot systems that first reached the clinic in the mid to late 1980s. Most of the drugs being released were potent molecules, like steroids, peptides and proteins. The pioneering work of Langer and Folkman, who showed proteins could be released from non-degradable polymer matrices, may have stimulated many people to think about other ways to deliver such drugs, such as by loading and releasing such drugs from biodegradable polymers. This history is described below.

In the 1960s, Ed Schmitt and “Roco” Polestina at Davis & Geck, Cyanamid Co. synthesized and patented poly(glycolic acid) (PGA) for use as a degradable suture [7]. Ethicon added lactic acid to the composition

to make the degradable poly(lactic-co-glycolic acid), PLGA, licensed the PGA technology from Davis and Geck, and brought out the degradable PLGA suture (Vicryl®). Kureha Co. in Japan developed the glycolide cyclic ester that made polymerization of glycolic acid practical, leading to the efficient production of PLA and PLGA.

In the late 1960s, Du Pont researchers, George Boswell and Richard Scribner added peptide drugs to PLA and fabricated microparticle and pellet depot DDS from them. They patented this technology in 1976, and it is interesting to note that they filed the patent application in 1969 [8]. (Fig. 6) In parallel, in the 1970s, researchers at Southern Research Institute (“Southern”) (e.g., Don Cowser and Danny Lewis) in collaboration with researchers at the University of Alabama at Birmingham (UAB) (e.g., Lee Beck) were developing and clinically testing steroid-loaded PLGA microparticles for contraceptive drug delivery; their work was supported by the US government (Program for Applied Research in Fertility Regulation, or PARFR, of the USAID) [9]. In the late 1970s, under a Southern project sponsored by Syntex, Tom Tice and Danny Lewis at Southern, and Lynda Saunders and John Kent at Syntex, developed and patented long-acting (1-month) PLGA microparticles containing leutinizing hormone-releasing hormone (LHRH) [10]. During the early 1980s, Southern developed a [D-Trp-6 LHRH]/PLGA microparticle for DebioPharm as a treatment of prostate cancer. This product, called Decapeptyl® LP, was launched in the clinic in Europe in 1986. It was the first clinically-approved, injectable, degradable microparticle depot DDS and is still on the market today [9]. Three years later, the Syntex patent developed at Southern in collaboration with, and assigned to Syntex [10] was licensed from Syntex by the Takeda-Abbott Pharmaceutical Co. (TAP) and marketed as the Lupron® depot [9].

Through the 1970s and 1980s, most of the degradable, drug-loaded microparticles were formed either by solvent evaporation from

United States Patent Office

3,773,919

Patented Nov. 20, 1973

1

2

3,773,919

POLYLACTIDE-DRUG MIXTURES

George Albert Boswell and Richard M. Scribner, Wilmington, Del., assignors to E. I. du Pont de Nemours and Company, Wilmington, Del.

No Drawing. Continuation-in-part of abandoned application Ser. No. 868,899, Oct. 23, 1969. This application Oct. 8, 1970, Ser. No. 79,309

Int. Cl. A61k 27/12

U.S. Cl. 424—19

16 Claims 10

ABSTRACT OF THE DISCLOSURE

Described and claimed are formulations of polylactide and drug to be introduced into the body which provide a slow sustained release of the drug over a controlled period of time. The polylactide is biodegradable in the body into normal or essentially normal metabolic products that have no deleterious or untoward effect on the body.

Description of the invention

The formulations of this invention are absorbable, non-irritating pharmaceutical compositions consisting of one or more drugs intimately mixed with or coated by a synthetic polylactide designed to release effective amounts of a drug over a predetermined period of time. The invention is of particular value for drugs that require prolonged administration or slow sustained release, for example certain fertility-control drugs or hormones used for hormone-replacement therapy. The polylactide may be considered as a carrier or matrix for the drug and in this document

the term "polylactide" includes both its generic meaning as a polyester derived from an α -hydroxycarboxylic acid and its specific meaning for the polymer derived from lactic acid (α -hydroxypropionic acid). The particular meaning in any given case will be apparent to one skilled in the art.

The novel formulations permit prolonged release of drugs for a controlled period of time from the sites of parenteral administration and minimize the frequency and thus the discomfort and inconvenience associated with conventional injection formulations. Unlike conventional depot injections, the formulations of this invention undergo biodegradation in the body into normal or essentially normal metabolic products, are nonreactive toward body tissue, and can be designed, by controlling molecular weight and composition, to undergo hydrolysis and to release drug from the depot at a desired rate.

Fig. 6. One of the earliest (if not the earliest) patents on drug-loaded degradable microparticles based on poly(lactic acid) microparticles synthesized at du Pont Chem. Co. [7].

United States Patent [19]

Gombotz et al.

[11] Patent Number: 5,019,400

[45] Date of Patent: May 28, 1991

[54] VERY LOW TEMPERATURE CASTING OF CONTROLLED RELEASE MICROSPHERES

[75] Inventors: Wayne R. Gombotz, Lexington; Michael S. Healy, E. Bridgewater; Larry R. Brown, Newton, all of Mass.

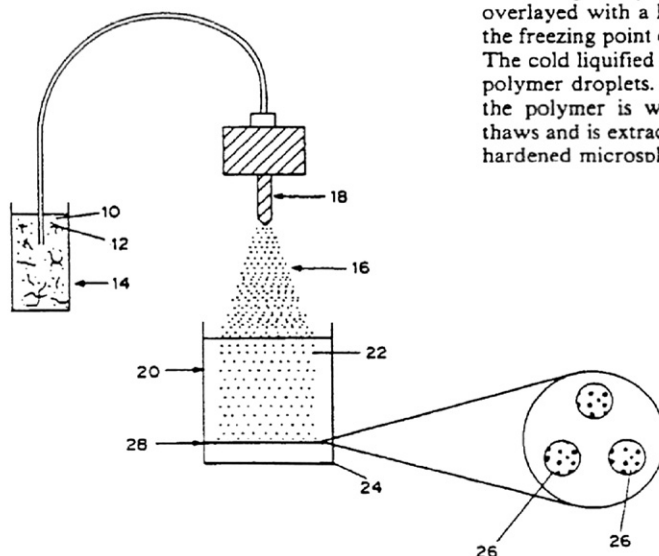
[73] Assignee: Enzytech, Inc., Cambridge, Mass.

[21] Appl. No.: 346,143

[22] Filed: May 1, 1989

[57] **ABSTRACT**

A process for preparing microspheres using very cold temperatures to freeze polymer-biologically active agent mixtures into polymeric microspheres with very high retention of biological activity and material. Polymer is dissolved in a solvent together with an active agent that can be either dissolved in the solvent or dispersed in the solvent in the form of microparticles. The polymer/active agent mixture is atomized into a vessel containing a liquid non-solvent, alone or frozen and overlaid with a liquified gas, at a temperature below the freezing point of the polymer/active agent solution. The cold liquified gas or liquid immediately freezes the polymer droplets. As the droplets and non-solvent for the polymer is warmed, the solvent in the droplets thaws and is extracted into the non-solvent, resulting in hardened microspheres.



Gombotz

Fig. 7. Drug-loaded PLGA microparticles were prepared in a process known as the "Prolease[®]" process, which was invented and patented by Gombotz, Healy and Brown, and assigned to Enzytech, Inc., a company founded by Bob Langer [11].

emulsions or by phase separation techniques. In the early 1990s Wayne Gombotz and others at Enzytech, a Langer spin-off company, invented the Prolease® process for fabricating uniform-sized, drug-loaded PLGA microparticles, which used an ultrasonic sprayer and a liquid nitrogen and frozen ethanol bath to freeze the uniform-sized particles and remove the solvent [11]. (Fig. 7) This process has been very popular for developing new drug-loaded, PLGA-based microparticle DDS. Alkermes bought Enzytech, and now owns the rights to the Prolease® process.

Another biodegradable polyester based on a copolymer of poly(ethylene glycol terephthalate) (PEG-T) and poly(butylene terephthalate) (PBT) was first synthesized and studied by Jan Feijen at Twente University in The Netherlands in the 1990s. Microparticle depot formulations of this polymer are now in clinical trials for delivery of α -Interferon (the product is called “Locteron®”, and is being developed by the Dutch company, Octopus).

In the late 1980s, Richard Dunn and others at Southern developed injectable, degradable drug depot DDS of PLA or PLGA; these depot are phase-separated, macroscopic “masses” formed by sub-cutaneous or IM injections of drug/polymer/solvent systems. (Dunn later moved to Atrix, Inc. where he extended this drug depot into the clinic). Other degradable depot systems, such as thermally-responsive, aqueous solutions of di-block and tri-block copolymers of PLGA-PEG, were developed by Sung Wan Kim and Youngro Byun at Utah University; they have been in clinical trials in Europe and were known as the “Re-Gel®” of MacroMed Inc. Alza developed a macroscopic depot DDS based on a degradable polyorthoester (POE) called “Chronomer®”, invented by Nam Choi and Jorge Heller at Alza. Heller has continued to develop injectable POE depot systems elsewhere, for use in the clinic. Langer’s lab (e.g., Bob Langer and his students and postdocs: Howard Rosen, Avi Domb, Kam Leong, Mark Salzman and others) produced a family of polyanhydrides that have seen clinical success in the hands of the surgeon, Henry Brem, MD, at Johns Hopkins Medical School, when loaded with drugs and implanted as macroscopic, disc-shaped depot systems for treatment of brain tumors (glioblastomas) [12].

Fig. 8 shows two of the key pioneers in the controlled DD field, Bob Langer at MIT and Sung Wan Kim the University of Utah. Both Langer and Kim have had immense influence on the growth of the controlled DD field and each has founded companies in this field that have led to clinical applications.

Inhalation of microscopic drug particles has been approved for a number of clinical uses. Pioneers in this area include John Patton a

Three Key Technologies that Stimulated Development of Nano-Carriers for Controlled Drug Delivery

1. PEGylation
2. Active targeting with antibodies, peptides and small molecule, cell-specific ligands
3. The EPR effect and passive targeting

Fig. 9. The three key technologies that stimulated the development of polymeric and nano-sized carriers as practical clinical realities.

founder of Inhale (later renamed Nektar) and more recently David Edwards (along with Langer) at MIT, who developed “flakey”, leaf-like, drug-loaded particles for inhalation therapies; based on this technology, they formed the company, AIR, which is now part of Alkermes.

4. The “NANO era” of targeted or site-controlled drug delivery systems

In the mid to late 1970s the concept of polymer-drug conjugates or “nano-therapeutics”, independently arose at various places around the world. Three key technologies were the major factors that stimulated the immense activity and clinical success of nano-therapeutics from the late 1980s to the present. (Fig. 9) The first was the concept of “PEGylation”, which refers to polyethylene glycol-conjugated drugs or drug carriers. The second is the concept of “active targeting” of the drug conjugate by conjugating cell membrane receptor antibodies, peptides or small molecule cell ligands to the polymer carrier. The third was the discovery of the “enhanced permeation and retention effect” (EPR) by Hiroshi Maeda in Kumamoto, Japan, wherein nano-scale carriers are entrapped within solid tumors due to leaky vasculature of the fast-growing tumor. This is called “passive” targeting as contrasted with active targeting. These will be discussed and referenced below.

The first major success of polymer-drug conjugates was based on the conjugation of poly(ethylene glycol) or PEG to the drug, known by the term “PEGylation”. In the late 1960s, Frank Davis at Rutgers



Bob Langer

Sung Wan Kim

Fig. 8. Bob Langer and Sung Wan Kim are two of the most significant contributors to the field of controlled drug delivery, both in conceiving and developing new DDS and also in translating those developments to industry and to the clinic. (Langer’s photo is by C.Bachrach Photography).



Fig. 10. PEGylation of drugs and their carriers was a critical element in the success of nano-scale DDS. The photo at the left (a) shows Frank Davis (left) and his PhD. student, Aubuchowski (right) at the opening of the first PEGylation company, Enzon, in the early 1980s. Davis had the original concept of PEGylation of protein drugs in the late 1960s. (source: Google images. The person in the middle is unidentified). The photo on the right (b) shows Milton Harris, the founder of the other major PEGylation company, Shearwater Polymers, speaking at the opening of the Shearwater Polymer Discovery Drive facility in Huntsville, AL in the 1990s. (Photo courtesy of Milton Harris). (See text for details).

University conceived of the concept of PEGylation to enhance both the circulation time and the stability (against enzyme attack or immunogenic recognition) of the recombinant protein drugs that were just being developed [13]. This led to the founding of the PEGylation company called Enzon, at the beginning of the 1980s. (Fig. 10a) The first clinical products were PEGylated enzymes such as asparaginase and glutaminase, which metabolized asparagine and glutamine, essential nutrients for leukemic cancer cells. Milton Harris, a chemistry professor at the University of Alabama, Huntsville later founded Shearwater Polymers, the other important PEGylation company, that subsequently collaborated with major pharmaceutical companies to introduce a number of PEGylated recombinant protein products to the clinic. (Fig. 10b)

Independent of Davis and around that same time in the 1970s, Helmut Ringsdorf at the University of Mainz sketched the idea of a targeted, polymer-drug conjugate and published it in 1975 [14]. (Fig. 11) Independent of Ringsdorf, Jindra Kopecek in Prague conceived of a new polymer carrier called poly(hydroxypropyl methacrylamide) (PHPMA) which was first synthesized by Karel Ulbrich, his PhD student; the drug was conjugated to the PHPMA by pendant tetrapeptide linkages that were degradable by cathepsin B, a lysosomal enzyme [15–17]. (Fig. 12) The polymer synthesis and characterization was carried out in Prague and the conjugate's drug action was tested in collaboration with Ruth Duncan, John Lloyd's PhD student, in the UK (Kopecek was introduced to Lloyd and Duncan by Ringsdorf). Duncan also contributed to the design of the polymer. Blanka Rihova in Prague found PHPMA to be non-immunogenic, and James Cassidy, MD, a UK clinician, led the clinical trials. The drugs included doxorubicin and other small molecule anti-cancer drugs. The drug-polymer conjugates could be actively targeted with ligands such as galactose, an asialo-glycoprotein membrane receptor ligand for

hepatocytes, for liver cancer treatment. Etienne Schacht of Ghent later synthesized new degradable peptide sequences. This was truly an international success story of a remarkable team of scientists and clinicians, bringing a novel polymer-drug conjugate to the clinic. (Fig. 12) This success has had a great influence on the field of nano-scale polymeric therapeutics. Duncan has published several reviews of nano-carriers and nano-therapeutics, one along with Kopecek [18,19].

Other polymer-small drug conjugates are currently being developed; examples include: Cell Therapeutics in Seattle with Xyotax®, a polyglutamic acid-paclitaxel conjugate, in phase III trials, and Insert Therapeutics of Mark Davis, with IT-101, a PEG-cyclodextrin-camptothecin polymeric micelle in phase II trials. Polymeric micelle-small drug and nucleic acid DDS will be discussed in more detail below.

One of the earliest examples of active targeting was the use of a polyclonal antibody to target a drug in the late 1950s [20]. The development and availability of monoclonal antibodies in the 1960s made it possible to deliver nano-therapeutics to specific cells. Other ligands have been discovered and have been used to target cells [21]. One of the most notable has been the integrin receptor ligand, the peptide RGD, first published in 1980 in *Science* by Pierschbacher and Ruoslahti [22].

In 1984, Hiroshi Maeda of Kumamoto University discovered what he called the “Enhanced Permeation and Retention” effect, or EPR. (Fig. 13) He was carrying out animal studies with his novel polymer-drug conjugate, styrene-maleic anhydride (SMA) conjugated to the anti-cancer peptide drug, neocarzinostatin (NCS), which he called “SMANCS” and he had labeled the conjugate with a dye. He noted that the dye accumulated within the tumor tissue, and concluded that the rapidly forming vasculature in such solid tumors was “leaky”, while the lymph drainage system was not yet working efficiently, and that led to its entrapment or accumulation within the tumor tissue. This

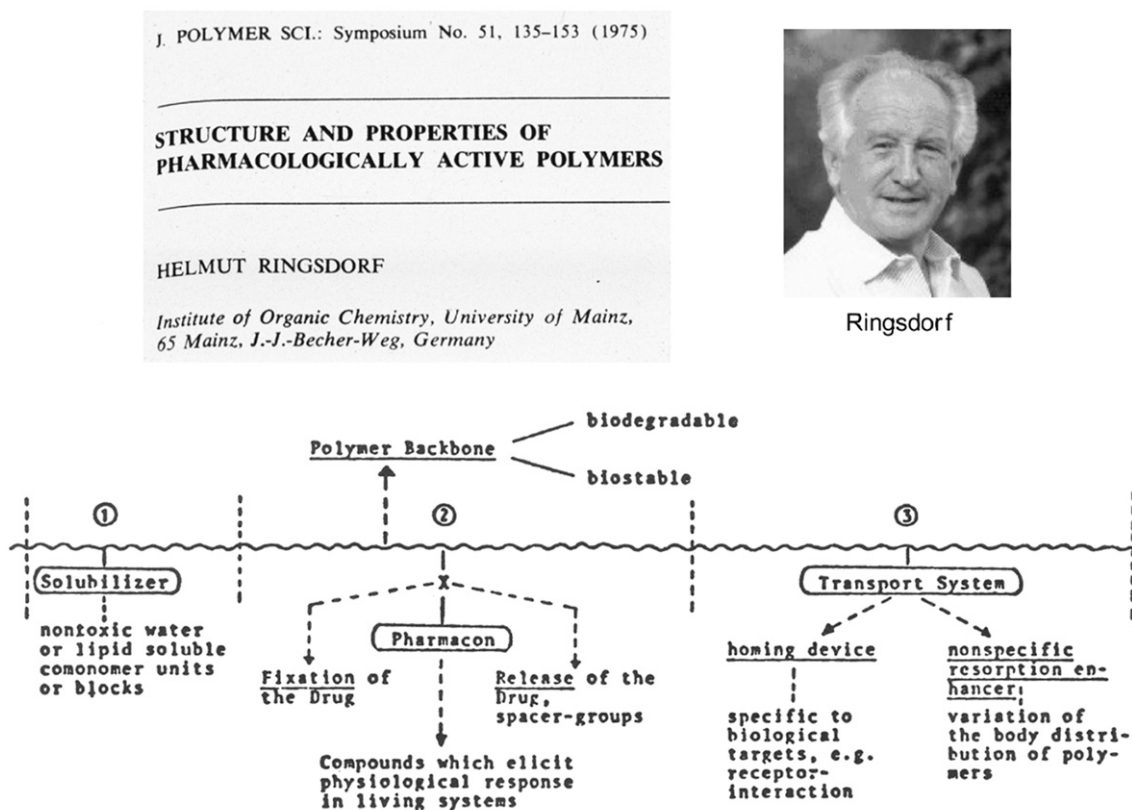


Fig. 11. An article by Helmut Ringsdorf in 1975 who independently proposed the concept of the “complete” polymer-drug construct, including targeting.

Soluble biomedical polymers

JINDŘICH KOPEČEK

Institute of Macromolecular Chemistry, Czechoslovak
Academy of Sciences, 162 06 Prague 6, Czechoslovakia

The first part of the paper is a review of literature data on the uses of synthetic soluble polymers in medicine. The second part is devoted to a discussion of the trend of investigation of medical polymers with a special attention to the development of new types of carriers of biologically active compounds.

This paper was presented at a conference on "Uses of Plastics in Medicine", Brno, Czechoslovakia, 1976, and published in "Polymers in Medicine", 1977, Vol VII, 191-219



Kopeček



Ulbrich



Kopečeková



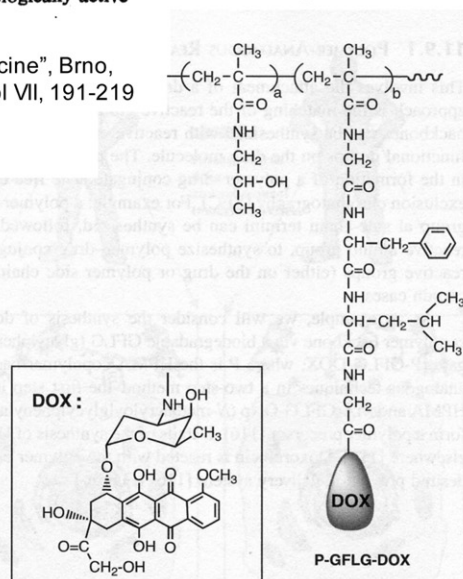
Rihová



Duncan



Cassidy



Cuchelkar & Kopeček in "Polymers in Drug Delivery", Eds Uchegbu & Schatzlein (2006) Taylor & Francis Publ., p 155-182

Fig. 12. Jindra Kopeček was an early pioneer who, in the mid-1970s, independently conceived of a practical polymer-drug conjugate with a releasable drug. Photos are shown of the international team that developed the first clinically successful synthetic polymer-drug conjugate having a degradable peptide linkage from the polymer to the drug. (See text for details).

combination caused the nano-scale SMANCS to be trapped within the extra-vascular tumor tissue. He submitted a manuscript on this observation, and Maeda recalls that Folkman was one of the reviewers; Folkman encouraged him to publish that exciting finding "as soon as possible" [23]. Recent evidence by various researchers suggests that the EPR effect is only effective close to the leaky vessels, and not throughout the tumor, due perhaps to the low diffusion coefficient of the nanocarriers within the tumor's extravascular tissues.

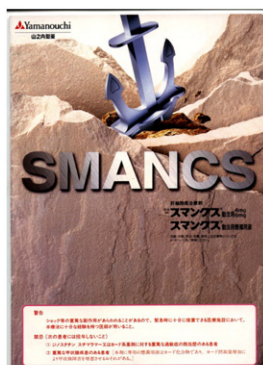
In the late 1980s and early 1990s other nano-scale DDS were developed, including PEGylated polymeric micelles and liposomes. Kazunori Kataoka, Teruo Okano and Masayuki Yokoyama in Tokyo synthesized A-B block copolymers of a PEG block conjugated to a hydrophobic amino acid block. These block copolymers spontaneously formed PEGylated polymeric micelles above a very low CMC. The hydrophobic cores of the micelles could be loaded with small hydrophobic drugs such as doxorubicin, either by physically loading the drug or by conjugating it to the amino acid pendant acid groups, and the terminal OH groups of the PEGs could be conjugated with cell specific ligands for targeted delivery [24–26]. Around essentially the same time, and independent of Kataoka et al.'s work, Alexander Kabanov in Nebraska developed drug-loaded PEGylated micelles based on PEO-PPO-PEO tri-block copolymers known as Pluronic® (where "PEO" = "PEG") [27]. (Fig. 14) Many different PEGylated polymeric micelles are now in clinical trials for delivery of a number of small molecule drugs.

In the early 1990s, the emergence of important nucleic acid drugs, such as plasmid DNA (pDNA) and antisense ODNs ("oligos") led to the development of cationically-condensed pDNA or ODN nanoparticles. Both cationic polymers and cationic liposomes were used to condense the nucleic acid drugs; the resultant complexes are called polyplexes or lipoplexes, respectively. One key polycation development was by Jean-Paul Behr, who proposed the use of poly(ethyleneimine) or PEI for complexation and intracellular delivery of nucleic acid drugs, where endosomal escape was enhanced by the PEI due to the "proton sponge" mechanism [28].

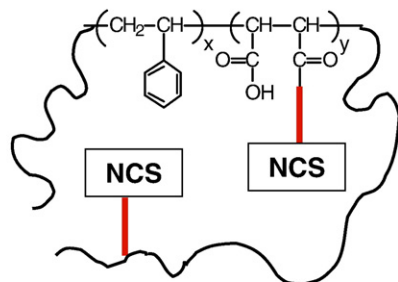
Block polymers of PEG-polycation (A-B) or PEG-polycation-PEG (A-B-A) have been used to condense a nucleic acid drug, to form PEGylated micelles, with the water insoluble nucleic acid-polycation electrostatic complexes (polyplexes) forming the core of the PEGylated polymeric micelle [29]. More recently, in the 2000s, a number of companies (e.g., Alnylam, Roche, Merck, Calando) have been involved in clinical trials for delivery of siRNA from similar lipoplexes and polyplexes. (e.g., [30])

Nano-scale albumin-based drug carriers have recently reached the clinic. Examples include Abraxane®, a nanoparticle of albumin and paclitaxel [31], and Albuferon-α®, a conjugate of albumin and interferon-α.

During the 1990s Vladimir Torchilin developed many liposomal formulations, some for diagnostic imaging applications and others for drug delivery, where hydrophilic drugs could be loaded in the aqueous core of the liposome, or hydrophobic drugs could be loaded in the lipid



Hiroshi Maeda



Styrene-Maleic Anhydride (SMA) conjugated to
NeoCarcinoStatin (NCS) ("SMANCS")
for treating liver cancer

Fig. 13. Hiroshi Maeda synthesized the polymer-drug conjugate of poly(styrene-co-maleic anhydride) (SMA) and NeoCarcinoStatin (NCS) and called it SMANCS. He discovered the EPR effect with this conjugate (See text for details).

bilayer shell [32]. A PEGylated liposome-doxorubicin product called Doxil® was approved by the FDA for clinical use in 1995. Martin Woodle and Frank Martin developed this product at Liposome Technologies Inc., (LTI). (Fig. 15)

Nano-scale DDS with polymeric carriers that are still under development include dendrimers, dendronized polymers and other hyper-branched polymers.

5. "Surface-controlled" drug delivery systems

Surface release of drugs from thin, drug-containing films or release of drugs from coated oral tablets were among the earliest surface-controlled DDS, which is another form of controlled drug delivery. These DDS have been clinically successful since the 1960s. These examples could also be categorized into one of, or a combination of, the macroscopic, microscopic and nanoscopic categories.

Sustained release of the anti-coagulant, heparin from polymer surfaces was one of the first successful drug delivery systems ever used in the clinic. This concept was brought into the clinic in the early 1960s by two clinicians, Vincent Gott, MD and James Whiffen, MD. It was based on an ionic complex of heparin, a highly negative polysaccharide, with a cationic surfactant that was hydrophobically-imbedded in the surface of the polymer being "heparinized", such as plastic or rubber tubing for carrying blood. The heparin was gradually eluted from the surface into the blood by ion exchange with the salt ions in the blood. It prevented coagulation of blood at the surface of the polymer until it reached a very low heparin concentration, usually after several days. ([33] is a review of heparinized surfaces).

A more recent clinical application of drug-loaded surface coatings is the drug-eluting stent (DES). (e.g., [34]) Two examples are the

Cypher® stent of Johnson & Johnson, and the Taxus® stent of Boston Scientific; each stent is designed to release a smooth muscle cell (SMC) antagonist (rapamycin or paclitaxel, respectively) that retards or prevents SMC proliferation, the main cause of vessel stenosis. Drug particles are blended with the polymer, which in the Cypher® stent is a polyblend of poly(butyl methacrylate) and polyEVA, while that used in the Taxus® stent is a novel tri-block copolymer of polystyrene-polyisobutylene-polystyrene (SIBS) developed by Joseph ("J.P.") Kennedy of the University of Akron. Other key contributors to the development of DES have been Mike Helmus of Boston Scientific, Len Pinchuk, formerly of J&J, and Art Coury, formerly of Medtronic. The DES are designed as matrix DDS, and a thin coating of pure polymer over the polymer-drug blend would convert them to reservoir devices.

There are other coatings that provided site-control of drug delivery, such as enteric coatings, developed in Europe by Rohm & Haas in the late 1950s. The earliest coatings were made of pH-responsive acrylic acid/ester copolymers (Eudragit®), and later, as developed by others, phthalic acid/ester copolymers of cellulose (CAP® of FMC Corp.). They were designed to be hydrophobic and non-swelling, with carboxyl groups protonated in the highly acidic gastric region, while in the enteric region, at pHs around 7–8, the carboxylic groups would be ionized, and the copolymer would swell and dissolve, releasing the drug.

Acrylic acid polymers have also been developed and clinically applied as site-controlled, mucoadhesive drug delivery systems with increased retention times at the surfaces of mucosal tissues such as the eye, nose, vagina and within the G-I tract, usually the stomach. Joe Robinson, Kinam Park and Nicholas Peppas have been major pioneers in this area. Lisbeth Illum has developed and clinically-applied mucoadhesive chitosan formulations for nasal delivery. Kinam Park has contributed to the related clinical development of gastric retention formulations based on mucoadhesive DDS that float as they become porous upon swelling and hydrating in the stomach. (e.g., [35])

6. The future of "controlled" drug delivery

There are many challenges and much excitement to come in the future of controlled DDS. As our knowledge of biology (especially cell biology and DNA) increases, so will our ability to design nano-scale DDS that are serum stable and efficiently taken up by specific cells, then escape the endosome and target specific sites and pathways within the cells. With this increased ability to control the efficiency and specificity of the delivery process, along with increased ability to design potent biomolecular drugs with minimal side effects, the field of controlled DDS will become ever more biological and less materials-oriented in character. Further, as our understanding continues to increase of which DNA sequences encode for which diseases, and then which sequences in the same individual DNA may be used to predict precise therapeutic regimes for optimum treatment of those diseases in each individual. Such "personalized medicine" will place demands on the drug delivery scientist to be more biologically precise and accurate with our "controlled" delivery systems. In the end (or is it really another beginning?) it will be even more exciting as the Controlled Drug Delivery field goes from the MACRO-MICRO-NANO POLYMERIC DDS eras to the BIO-DDS era of *bio-controlled delivery systems for biomolecular drugs*.

Acknowledgments

Much of the information in this article has been the author's personal knowledge, gained over the past 40+ years, and in addition, the author has solicited personal information from the many individuals involved in the history and evolution of the controlled drug delivery field, and Jindra Kopecek, Tom Tice and Howie Rosen have been especially helpful. It is worth noting one more time that this

The Polymeric Micelle as a Drug Carrier

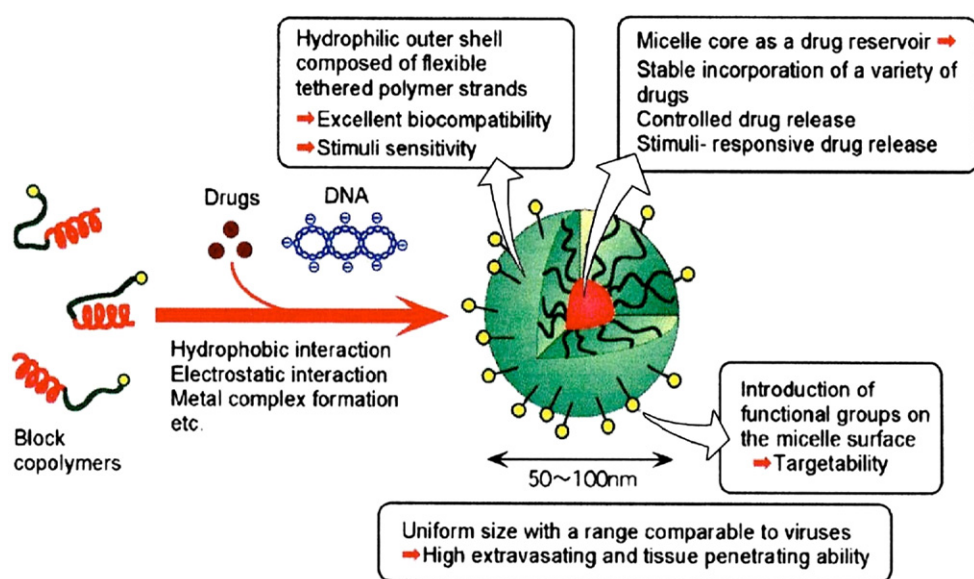


Fig. 1. Polymeric micelles as intelligent nanocarriers for drug and gene delivery.

Pluronic®
micelles



Kabanov

PEGylated
micelles



Kataoka

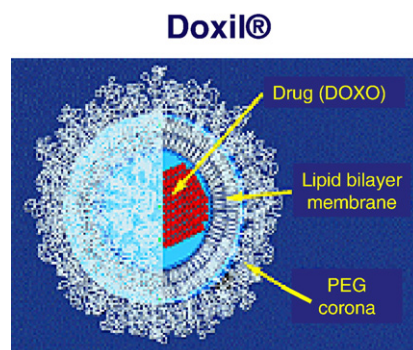


Okano



Yokoyama

Fig. 14. The PEGylated polymeric micelle drug carrier was developed in Japan by Kazunori Kataoka, Teruo Okano and Masayuki Yokoyama in the late 1980s and early 1990s. A polymeric micelle stabilized by Pluronic® polyols was independently developed by Alexander Kabanov in the US around the same time. These drug-carrying micelles are currently in clinical trials. (See text for more details).

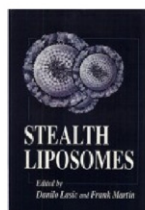


Doxil® was FDA-approved in 1995

http://www.doxil.com/images/mm/upper_doxil_logo.jpg



Martin Woodle
(with Frank Martin
at Liposome
Technology, Inc.)



CRC Press 1995

article is about the early history and evolution of *clinical applications* of controlled drug delivery (CDD). If it were a survey of the immense scientific literature of CDD, the contributions of many others, especially Nicholas Peppas, would have been highlighted. In addition, Google and the Internet have been highly valuable sources of information and images, including many photos. For these reasons, the reference list is limited to a relatively few literature citations. Many of the figures were taken from old photos and slides, and their poor quality is the responsibility of the author, and not that of the editors or the publisher. The author is grateful that he was allowed to include those fuzzy figures in this article.

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Fig. 15. A PEGylated liposome containing the anti-cancer drug Doxorubicin was developed in the early 1990s at Liposome Technology, Inc. by Martin Woodle and Frank Martin. It is called "Doxil®" and was approved for clinical use in 1995. The PEGylated carrier was called "stealth" because of its long circulating characteristic.

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